MATHEMATICS

University of NORTH ALABAMA

Spring 2025 Math Problem of the Week: Problem #12

Due noon Monday, April 14
Open to all UNA undergraduates

Directions: Deliver your written solution to me (Dr. Bryant) in CPTMTH 328 by noon Monday, April 14 or, if you cannot find me, you may leave it with Ms. Buckley in CPTMTH 377. All work must be your own. You are required to justify your response fully. If you utilize a well-known theorem, state the theorem's full name and proposition, showing that the conditions of the theorem are met. Good luck!

Problem #12: A heavy homogeneous cable of length l, which is fastened at its upper end $x = 0$ to a vertical axis, rotates about this axis with a uniform angular velocity ω. Derive the differential equation for small transverse vibrations in the neighborhood of the vertical state of equilibrium.

N.B. the cable is of such substantial weight that under the velocity ω it does not leave a small neighborhood of the vertical axis.